

Documentation for Precision Beekeeping Ontology (PBO)

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May 2026

1 Introduction

This documentation contains technical information about the Precision Beekeeping Ontology (PBO) developed to represent knowledge in beekeeping domain, particularly precision beekeeping (PB). The PBO ontology targets the researchers, agriculturalists particularly beekeepers, software developers and ontology engineering community. The PBO ontology covers the following areas in hive and colony health management in precision beekeeping: observation such as temperature, weight and humidity observations that represent actual measurement events and its results. It covers observable property such

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as temperature, weight and humidity that represents what can be measured. It covers devices used in PB such as smart devices, hives, power supply system and network router. Also, it covers the monitoring system and alert about connectivity and power. It covers hive and monitoring system status. It covers about bee and their roles.

This documentation is organized as follows: Section 2 contains PBO ontology metadata; Section 3 gives information about reused external thesauri and ontologies; Section 4 is about PBO ontology structure; Section 5 explains about constraints on properties in PBO ontology; Section 6 talks about SPARQL queries and results of PBO ontology; Section 7 concludes the documentation.

2 PBO ontology metadata

- *Ontology name*: The name given to the developed ontology is Precision Beekeeping Ontology (PBO).
- *Version*: This is the first version (V1.0) of the PBO ontology.
- *Ontology availability*: The PBO ontology is available at <https://github.com/paulmushi0018/Precision-Beekeeping-Ontology-PBO->
- *License*: The PBO ontology is released under Creative Commons licenses (CC BY-SA)¹
- *Language and tools used*: The tool used to develop ontology is Protégé² 5.6.5, which supports OWL 2 (Web Ontology Language), Resource Description Framework (RDF) and Description Logic (DL). The SPARQL³ queries were done in Apache Jena Fuseki Server 5.4.0 and Logical consistency was checked using HermiT 1.4.3.456 reasoner in Protégé. Visualization tool, WebVOWL⁴ was used.

3 Reused external thesauri and ontologies

The PBO ontology reused the concepts and relations from DOLCE+DnS Ultralite (DUL) upper ontology⁵, SOSA Module (Sensors, Observations, Sam-

¹<https://creativecommons.org/licenses/by-sa/4.0/>

²<https://protege.stanford.edu/software.php#desktop-protege>

³https://sparql.dev/article/SPARQL_and_ontology_development.html

⁴<https://service.tib.eu/webvowl/>

⁵<http://www.ontologydesignpatterns.org/ont/dul/DUL.owl>

ples and Actuations) in Semantic Sensor network ontology (SSN)⁶, AGROVOC⁷ Multilingual Thesaurus and the General Multilingual Environmental Thesaurus (GEMET⁸), for example “c.8872” for “Apiary_” and all other classes defined in Protégé with URI in external vocabulary were annotated by *rdfs:label* for readability. This can be seen in Figure 1 for the term Apiary_ and Bee-Colony_ that they have external link to AGROVOC thesaurus.

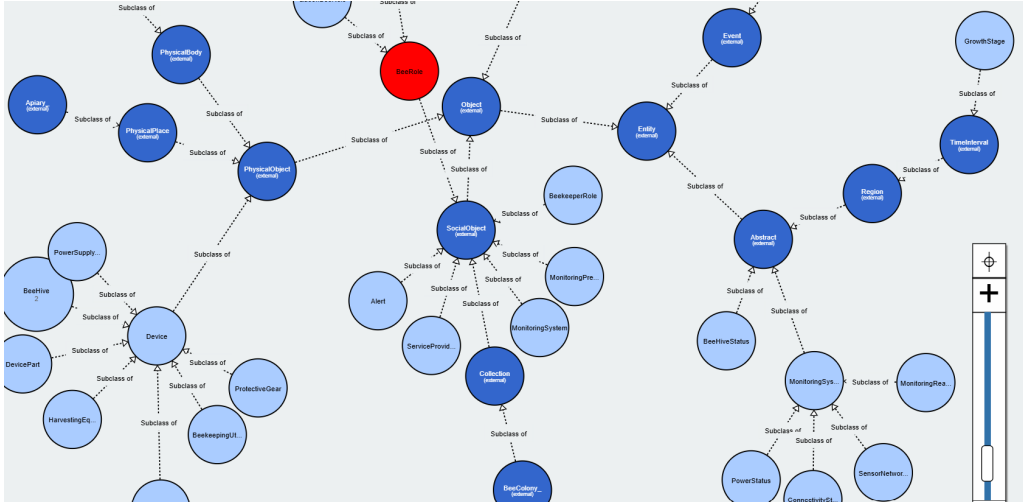


Figure 1: Class hierarchy and visualization of the PBO ontology using the WebVOWL tool

4 PBO ontology structure

As depicted in Figure 2, metrics of PBO ontology in Protégé are axioms, logical axioms count, declarative axioms count, class count, object property count, data property count, individual count and annotation property count.

- PBO ontology has 1097 axioms, whereby 707 axioms are logical and 360 axioms are declarative.
- PBO ontology has 189 concepts or terms that appear as classes in Protégé, which indicates the concepts that have been covered in the PB domain.

⁶<https://w3c.github.io/sdw-sosa-ssn/ssn/>

⁷https://agrovoc.fao.org/browse/agrovoc/en/page/c_529

⁸<https://www.eionet.europa.eu/gemet/en/themes/>

Ontology metrics	
Metrics	
Axiom	1,097
Logical axiom count	707
Declaration axioms count	360
Class count	189
Object property count	71
Data property count	26
Individual count	74
Annotation Property count	2
Class axioms	
SubClassOf	188
EquivalentClasses	0

Figure 2: Metrics of the PBO ontology from the Protégé

- The object property count in PBO ontology is 71, which defines the relationship between classes. For example, the object property that links the “Observation” class and the “BeeHive” class is shown in Figure 3 “hasFeatureOfInterest” which was reused from SSN/SOSA. Also, some of object properties are seen in Figure 4 as defined in the Protégé.

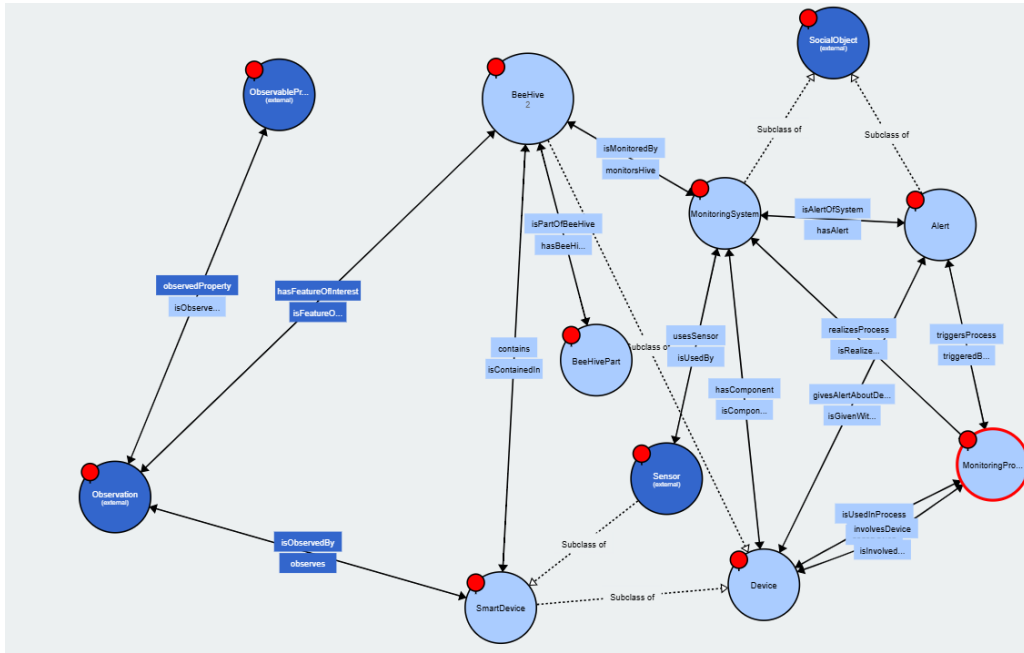


Figure 3: Visualization in WebVOWL tool of the object properties such as hasFeatureOfInterest between the Observation class and the BeeHive class.

- The data property count is 26 in the PBO ontology and some of them are seen in Figure 5.

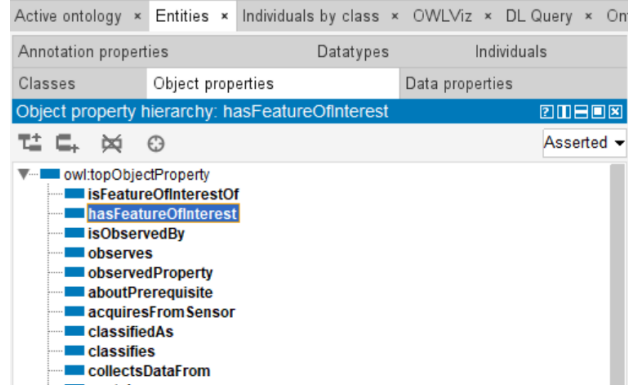


Figure 4: Snippet of Object Property Hierarchy in Protégé for PBO Ontology

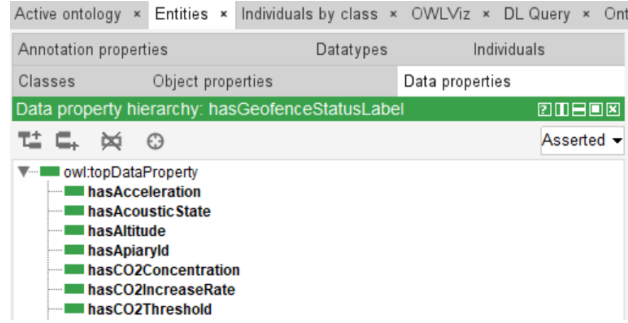


Figure 5: Snippet of Data Property Hierarchy in Protégé for PBO Ontology

- The individual count is 74 in PBO ontology as seen in Table 2 and Table 3, which contains about beehive metrics that include 2 hives (Schwartau and Würzburg), 48 internal temperature readings of Schwartau hive and 24 hive weight readings of Schwartau and Würzburg hives from the Kaggle dataset repository⁹ as seen in Figure 6.

5 Constraints on properties in PBO ontology

To ensure the correct reasoning of the PBO ontology, we specified constraints on the properties. For example, the value datatype constraints show that the *WeightObservation* class can contain individual *WeightObs1* specified under the data property *hasWeight* with value *50.55* as double datatype and not otherwise as seen in Figure 7.

We also specified the domain and range restrictions on the object prop-

⁹<https://www.kaggle.com/datasets/se18m502/bee-hive-metrics/data>

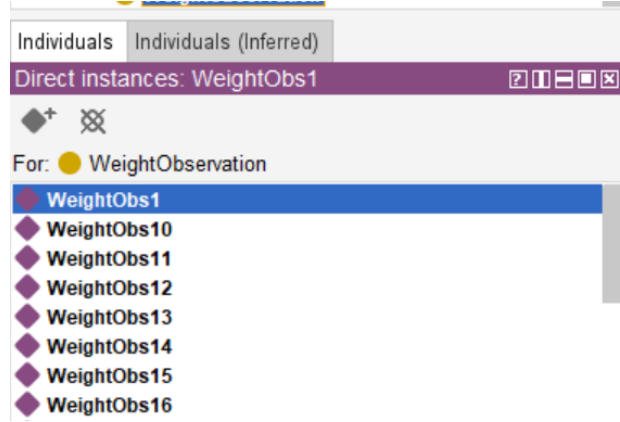


Figure 6: Snippet of individuals (instances) in Protégé for PBO Ontology



Figure 7: Snippet of data Property restrictions in Protégé for PBO Ontology

erties for example `hasFeatureOfInterest` object property with the domain `Observation` class and the range `BeeHive` class.

In specifying restrictions on data properties, we specified value datatypes, domain and range restrictions such as `hasWeight` data property link between `BeeHive` class as domain and `xsd:double` as range.

6 SPARQL queries and results of PBO ontology

The PBO ontology was evaluated based on the pre-defined competence questions (CQs) as seen in Table 1 that were 6 based on the internal temperature and hive weight data as seen in Table 3 and Table 2 respectively. Using SPARQL queries in the Apache Jena Fuseki server after being set locally as

shown in Figure 10. We successfully evaluated the ontology based on 6 CQs where CQ1, CQ2 and CQ3 are sensor-based on the weight observations of hives as seen in Figure 8 that shows the interplay of classes, object properties and data properties while in Figure 9 is the interplay of classes, object properties and data properties for sensor-based internal temperature observation to evaluates CQ4, CQ5 and CQ6.. Figure 11 shows the total number of triples uploaded 1098, from the PBO ontology. The Example queries 1 to 6 for predefined 6 CQs are discussed in this section and the implementation with the results.

Table 1: Executable competency questions and their evaluation intention

CQ#	Competence Question	Evaluation intention
CQ1	For each hive, what is its most recent recorded weight?	Tests retrieval of the latest hive-weight observation using hive, timestamp and weight-value data.
CQ2	On which days did each hive experience strong daily nectar intake, defined as a daily weight increase above a threshold?	Tests whether daily weight increases can be detected, indicating possible nectar intake, honey accumulation or strong foraging activity.
CQ3	On which days did each hive experience significant daily weight loss below a given threshold?	Tests whether relevant daily weight losses can be detected, indicating possible swarming, honey removal, food consumption or colony stress.
CQ4	Is the Schwartau hive maintaining a stable internal hive temperature?	Tests retrieval and interpretation of internal-temperature observations for assessing hive-temperature stability.
CQ5	Did the Schwartau hive experience rapid temperature drops greater than 2°C per hour, indicating weakening thermoregulation or disturbance?	Tests comparison of consecutive temperature observations to detect rapid internal-temperature drops that might be due to colony stress.
CQ6	Is the internal temperature showing a consistent upward or downward trend, indicating cluster migration?	Tests whether temporal temperature trends can be detected, indicating possible cluster movement or behavioural change.

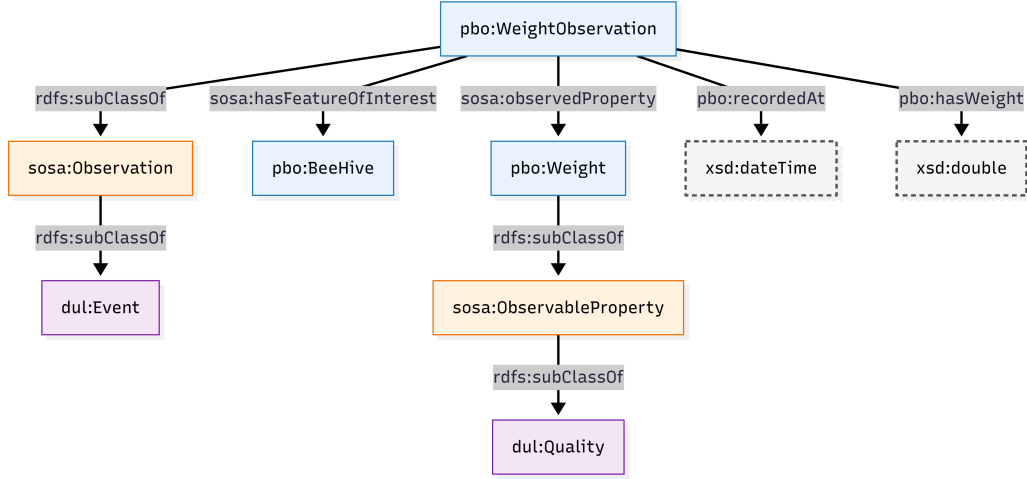


Figure 8: Interplay of classes, object properties and data properties for CQ1, CQ2 and CQ3 about sensor-based weight observations in hive

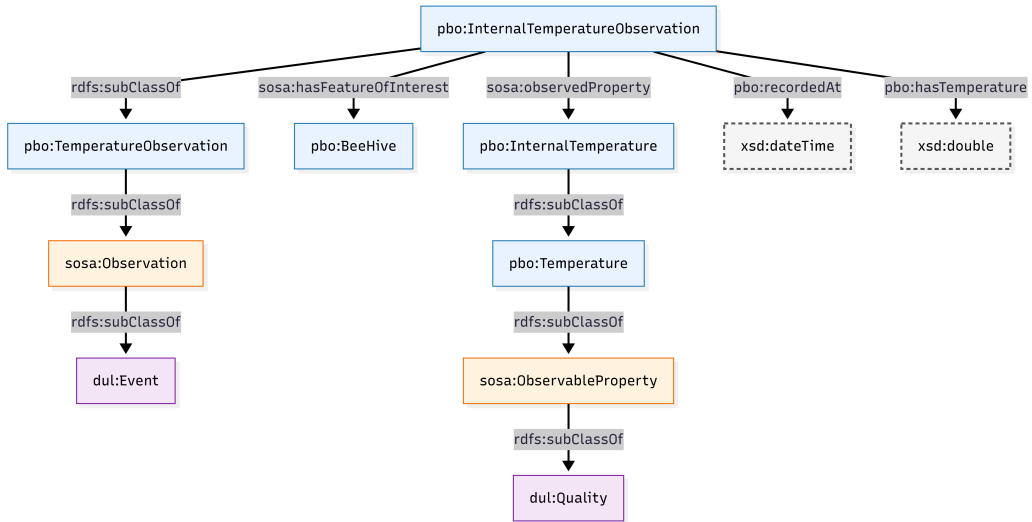


Figure 9: Interplay of classes, object properties and data properties for CQ4, CQ5 and CQ6 about sensor-based internal temperature observations in hive

```

Microsoft Windows [Version 10.0.26100.7462]
(c) Microsoft Corporation. All rights reserved.

C:\Users\paulk>java -version
java version "24.0.1" 2025-04-15
Java(TM) SE Runtime Environment (build 24.0.1+9-30)
Java HotSpot(TM) 64-Bit Server VM (build 24.0.1+9-30, mixed mode, sharing)

C:\Users\paulk>cd C:\apache-jena-fuseki-5.4.0

C:\apache-jena-fuseki-5.4.0>fuseki-server --update -mem /ds_pbo
15:03:20 INFO Server      :: Apache Jena Fuseki 5.4.0
15:03:20 INFO Config      :: Fuseki Base = C:\apache-jena-fuseki-5.4.0\run
15:03:20 INFO Config      :: Load configuration: file:///C:/apache-jena-fuseki-5.4.0/run/configuration/ds_ttl
15:03:20 INFO Config      :: Database: /ds_pbo
15:03:20 INFO Config      :: Database: /ds
15:03:20 INFO Config      :: UI Base = fuseki-server.jar
15:03:21 ERROR Server      :: Expected only one dataset in the DataAccessPointRegistry
15:03:21 INFO Server      :: Database: in-memory
15:03:21 INFO Server      :: Path = /ds
15:03:21 INFO Server      :: Path = /ds_pbo
15:03:21 INFO Server      :: Memory: 4.0 GiB
15:03:21 INFO Server      :: Java: 24.0.1
15:03:21 INFO Server      :: OS: Windows 11 10.0 amd64
15:03:21 INFO Server      :: PID: 16468
15:03:21 INFO Shiro        :: Shiro configuration: file:C:\apache-jena-fuseki-5.4.0\run\shiro.ini
15:03:21 INFO Server      :: Start Fuseki (http=3030)

```

Figure 10: Setting the Apache Jena Fuseki Server for Ontology Evaluation


Apache Jena Fuseki
datasets
manage
help
server status

[query](#)
[add data](#)
[edit](#)
[info](#)

/dataset_pbo

Upload files /dataset_pbo/data

Load data into the default graph of the currently selected dataset, or the given named graph. You may upload any RDF format, such as Turtle, RDF/XML or TRIG.

Dataset graph name

Files to upload

[+ select files](#)
[upload all](#)

name	size	speed	status	actions
revisedPBO.rdf	163.58kb	163.77kb/s	100.00 Triples uploaded: 1098	upload now remove

Figure 11: Triples uploaded in Apache Jena Fuseki server for running SPARQL queries

Table 2: Sample weight observations for Würzburg and Schwartau hives

Instance	Hive	Hive ID	Recorded Timestamp	Weight (kg)
WeightObs1	Würzburg	Würzburg1	2017-01-01T13:00:00	52.63
WeightObs2	Würzburg	Würzburg1	2017-01-02T01:00:00	52.67
WeightObs3	Würzburg	Würzburg1	2017-01-02T13:00:00	52.70
WeightObs4	Würzburg	Würzburg1	2017-01-03T01:00:00	52.70
WeightObs5	Würzburg	Würzburg1	2017-01-03T13:00:00	52.63
WeightObs6	Würzburg	Würzburg1	2017-01-04T01:00:00	52.63
WeightObs7	Würzburg	Würzburg1	2017-01-04T13:00:00	52.67
WeightObs8	Würzburg	Würzburg1	2017-01-05T01:00:00	52.60
WeightObs9	Würzburg	Würzburg1	2017-01-05T13:00:00	52.57
WeightObs10	Würzburg	Würzburg1	2017-01-06T01:00:00	52.67
WeightObs11	Würzburg	Würzburg1	2017-01-06T13:00:00	52.89
WeightObs12	Würzburg	Würzburg1	2017-01-07T01:00:00	52.38
WeightObs13	Schwartau	Schwartau1	2017-01-01T13:00:00	50.74
WeightObs14	Schwartau	Schwartau1	2017-01-02T01:00:00	50.70
WeightObs15	Schwartau	Schwartau1	2017-01-02T13:00:00	50.61
WeightObs16	Schwartau	Schwartau1	2017-01-03T01:00:00	50.74
WeightObs17	Schwartau	Schwartau1	2017-01-03T13:00:00	50.80
WeightObs18	Schwartau	Schwartau1	2017-01-04T01:00:00	50.65
WeightObs19	Schwartau	Schwartau1	2017-01-04T13:00:00	50.41
WeightObs20	Schwartau	Schwartau1	2017-01-05T01:00:00	50.17
WeightObs21	Schwartau	Schwartau1	2017-01-05T13:00:00	50.05
WeightObs22	Schwartau	Schwartau1	2017-01-06T01:00:00	49.88
WeightObs23	Schwartau	Schwartau1	2017-01-06T13:00:00	50.46
WeightObs24	Schwartau	Schwartau1	2017-01-07T01:00:00	50.55

Table 3: Sample internal temperature observations for the Schwartau hive

Instance	Hive	Hive ID	Recorded Timestamp	Internal Temperature (°C)
InTemperatureObs1	Schwartau	Schwartau1	2017-01-01T14:15:00	12.34
InTemperatureObs2	Schwartau	Schwartau1	2017-01-01T15:15:00	12.38
InTemperatureObs3	Schwartau	Schwartau1	2017-01-01T16:15:00	15.73
InTemperatureObs4	Schwartau	Schwartau1	2017-01-01T17:15:00	17.63
InTemperatureObs5	Schwartau	Schwartau1	2017-01-01T18:15:00	13.35
InTemperatureObs6	Schwartau	Schwartau1	2017-01-01T19:15:00	13.93
InTemperatureObs7	Schwartau	Schwartau1	2017-01-01T20:15:00	12.52
InTemperatureObs8	Schwartau	Schwartau1	2017-01-01T21:15:00	20.56
InTemperatureObs9	Schwartau	Schwartau1	2017-01-01T22:15:00	19.91
InTemperatureObs10	Schwartau	Schwartau1	2017-01-01T23:15:00	19.99
InTemperatureObs11	Schwartau	Schwartau1	2017-01-02T00:15:00	18.24
InTemperatureObs12	Schwartau	Schwartau1	2017-01-02T01:15:00	13.47
InTemperatureObs13	Schwartau	Schwartau1	2017-01-02T02:15:00	13.20
InTemperatureObs14	Schwartau	Schwartau1	2017-01-02T03:15:00	14.86
InTemperatureObs15	Schwartau	Schwartau1	2017-01-02T04:15:00	12.34
InTemperatureObs16	Schwartau	Schwartau1	2017-01-02T05:15:00	12.77
InTemperatureObs17	Schwartau	Schwartau1	2017-01-02T06:15:00	13.90
InTemperatureObs18	Schwartau	Schwartau1	2017-01-02T07:15:00	15.16
InTemperatureObs19	Schwartau	Schwartau1	2017-01-02T08:15:00	11.72
InTemperatureObs20	Schwartau	Schwartau1	2017-01-02T09:15:00	12.75
InTemperatureObs21	Schwartau	Schwartau1	2017-01-02T10:15:00	12.52
InTemperatureObs22	Schwartau	Schwartau1	2017-01-02T11:15:00	11.50
InTemperatureObs23	Schwartau	Schwartau1	2017-01-02T12:15:00	12.92
InTemperatureObs24	Schwartau	Schwartau1	2017-01-02T13:15:00	14.60
InTemperatureObs25	Schwartau	Schwartau1	2017-01-02T14:15:00	11.91
InTemperatureObs26	Schwartau	Schwartau1	2017-01-02T15:15:00	12.16
InTemperatureObs27	Schwartau	Schwartau1	2017-01-02T16:15:00	11.63
InTemperatureObs28	Schwartau	Schwartau1	2017-01-02T17:15:00	11.43
InTemperatureObs29	Schwartau	Schwartau1	2017-01-02T18:15:00	11.03
InTemperatureObs30	Schwartau	Schwartau1	2017-01-02T19:15:00	10.60
InTemperatureObs31	Schwartau	Schwartau1	2017-01-02T20:15:00	11.67
InTemperatureObs32	Schwartau	Schwartau1	2017-01-02T21:15:00	13.48
InTemperatureObs33	Schwartau	Schwartau1	2017-01-02T22:15:00	20.08

Continued on next page

Instance	Hive	Hive ID	Recorded Timestamp	Internal Temperature (°C)
InTemperatureObs34	Schwartau	Schwartau1	2017-01-02T23:15:00	15.84
InTemperatureObs35	Schwartau	Schwartau1	2017-01-03T00:15:00	16.57
InTemperatureObs36	Schwartau	Schwartau1	2017-01-03T01:15:00	15.75
InTemperatureObs37	Schwartau	Schwartau1	2017-01-03T02:15:00	19.50
InTemperatureObs38	Schwartau	Schwartau1	2017-01-03T03:15:00	12.33
InTemperatureObs39	Schwartau	Schwartau1	2017-01-03T04:15:00	12.67
InTemperatureObs40	Schwartau	Schwartau1	2017-01-03T05:15:00	11.43
InTemperatureObs41	Schwartau	Schwartau1	2017-01-03T06:15:00	14.14
InTemperatureObs42	Schwartau	Schwartau1	2017-01-03T07:15:00	11.00
InTemperatureObs43	Schwartau	Schwartau1	2017-01-03T08:15:00	15.28
InTemperatureObs44	Schwartau	Schwartau1	2017-01-03T09:15:00	15.49
InTemperatureObs45	Schwartau	Schwartau1	2017-01-03T10:15:00	15.24
InTemperatureObs46	Schwartau	Schwartau1	2017-01-02T11:15:00	15.62
InTemperatureObs47	Schwartau	Schwartau1	2017-01-02T12:15:00	20.91
InTemperatureObs48	Schwartau	Schwartau1	2017-01-02T13:15:00	15.63

6.1 Example Query 1 for CQ1

The query as seen in Listing 1 uses instances of `pbo:WeightObservation` class to represent weight of hive measurements. Each weight observation is linked to the corresponding hive through the object property `sosa:hasFeatureOfInterest`. The timestamp of each observation is represented by the data property `pbo:recordedAt`, while the value of hive’s weight is represented by the data property `pbo:hasWeight`.

The query first groups the observations by hive and identifies the latest timestamp of each hive using `MAX(?t)`. It then retrieves the weight observation corresponding to that latest timestamp and returns the hive, time and recorded weight. This enables the developed PBO ontology to answer the question of the most recent available weight of each hive measurement.

As shown in Table 4, the query returned two latest recorded weight per hive. These results demonstrate that the developed PBO ontology can support sensor-based retrieval of the latest hive status by connecting hive individuals, timestamps, weight observations and measured weight values through classes, object properties and data properties as seen in Figure 8.

Listing 1: SPARQL query for Competence Question 1 (CQ1) in Jena Fuseki

Server for Ontology Evaluation

```
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
PREFIX xsd: <http://www.w3.org/2001/XMLSchema#>
PREFIX pbo: <http://www.semanticweb.org/paulk/ontologies/2025/10/
    revisedPBO#>
PREFIX sosa: <http://www.w3.org/ns/sosa/>
SELECT ?hive ?time ?weight
WHERE {
  {
    SELECT ?hive (MAX(?t) AS ?time)
    WHERE {
      ?obs rdf:type pbo:WeightObservation ;
        sosa:hasFeatureOfInterest ?hive ;
        pbo:recordedAt ?t .
    }
    GROUP BY ?hive
  }
  ?obs rdf:type pbo:WeightObservation ;
    sosa:hasFeatureOfInterest ?hive ;
    pbo:recordedAt ?time ;
    pbo:hasWeight ?weight .
}
ORDER BY ?hive
```

Table 4: SPARQL query results for Competence Question 1 (CQ1): most recent recorded weight for each hive

Hive	Time	Weight (kg)
Schwartau	2017-01-07T01:00:00	50.55
Würzburg	2017-01-07T01:00:00	52.38

6.2 Example Query 2 for CQ2

Example Query 2 in Listing 2 evaluates whether each hive experienced a positive daily weight increase, which may indicate nectar intake, honey accumulation or strong foraging activity. The query uses instances of `pbo:WeightObservation` to represent hive-weight measurements. Each weight observation is linked to the corresponding hive through the object property `sosa:hasFeatureOfInterest`. The timestamp of each observation is represented by the data property `pbo:recordedAt`, while the measured hive-weight value is represented by the data property `pbo:hasWeight`.

The query first groups observations by hive and calendar day using BIND (`xsd:date(?t)AS?day`). For each hive and day, it identifies the first and last recorded timestamps using MIN(`?t`) and MAX(`?t`). It then retrieves the corresponding start and end weight values and computes the daily weight change as `?endWeight - ?startWeight`. The filter condition keeps only cases where the daily change is greater than zero, meaning that the hive weight increased during that day.

As shown in Table 5, the query returned five day cases of the hive where the end weight was higher than the start weight. These results demonstrate that the developed PBO ontology can support sensor-based evaluation of daily hive-weight changes by combining classes, object properties and data properties as seen in Figure 8. In particular, the ontology connects hive individuals, weight observations, timestamps and measured weight values in a way that enables SPARQL-based detection of days with positive weight gain.

Listing 2: SPARQL query for Competence Question 2 (CQ2) in Jena Fuseki Server for Ontology Evaluation

```
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
PREFIX xsd: <http://www.w3.org/2001/XMLSchema#>
PREFIX pbo: <http://www.semanticweb.org/paulk/ontologies/2025/10/
    revisedPBO#>
PREFIX sosa: <http://www.w3.org/ns/sosa/>
SELECT ?hive ?day ?startWeight ?endWeight (?endWeight - ?
    startWeight AS ?dailyChange)
WHERE {
    # 1) For each hive + day, get first and last timestamps
    {
        SELECT ?hive ?day (MIN(?t) AS ?startTime) (MAX(?t) AS ?endTime)
        WHERE {
            ?obs rdf:type pbo:WeightObservation ;
                sosa:hasFeatureOfInterest ?hive ;
                pbo:recordedAt ?t .
            # Optional: limit to a specific period
            FILTER(
                ?t >= "2017-01-01T00:00:00"^^xsd:dateTime &&
                ?t < "2017-01-31T00:00:00"^^xsd:dateTime
            )
            # Derive calendar day from timestamp
            BIND( xsd:date(?t) AS ?day )
        }
    }
    GROUP BY ?hive ?day
```

```

}
# 2) Get weight at startTime
?startObs rdf:type pbo:WeightObservation ;
          sosa:hasFeatureOfInterest ?hive ;
          pbo:recordedAt ?startTime ;
          pbo:hasWeight ?startWeight .

# 3) Get weight at endTime
?endObs   rdf:type pbo:WeightObservation ;
          sosa:hasFeatureOfInterest ?hive ;
          pbo:recordedAt ?endTime ;
          pbo:hasWeight ?endWeight .

# 4) Keep only significant gains (e.g. greater than 0 kg)
FILTER( (?endWeight - ?startWeight) > 0 )
}
ORDER BY ?hive ?day

```

Table 5: SPARQL query results for Competence Question 2 (CQ2): daily hive-weight increases

Hive	Day	Start weight (kg)	End weight (kg)	Daily change (kg)
Schwartau	2017-01-06	49.88	50.46	0.58
Schwartau	2017-01-03	50.74	50.80	0.06
Würzburg	2017-01-04	52.63	52.67	0.04
Würzburg	2017-01-02	52.67	52.70	0.03
Würzburg	2017-01-06	52.67	52.89	0.22

6.3 Example Query 3 for CQ3

Example query 3 evaluates whether a monitored hive experienced a significant daily weight loss within a given day as shown in Listing 3. The query uses instances of `pbo:WeightObservation` class to represent weight measurement of hive collected from sensors. Each observation is linked to the corresponding hive through the object property `sosa:hasFeatureOfInterest`. The timestamp of each observation is represented by the data property `pbo:recordedAt`, while the measured hive-weight value is represented by the data property `pbo:hasWeight`.

The query first groups observations by hive and calendar day using `BIND(xsd:date(?t)AS?day)`. For each hive and day, it identifies the first and last recorded timestamps using `MIN(?t)` and `MAX(?t)`. It then retrieves the corresponding start and end hive-weight values and computes the daily

weight change as $(?endWeight - ?startWeight)$. The filter condition keeps only cases where the daily change is less than zero, indicating that the hive weight decreased during that day.

As shown in Table 6, the query returned five records of the hive in a given day observations where the ending hive weight was lower than the starting hive weight. These results demonstrate that the developed PBO ontology can support sensor-based monitoring of weight decrease patterns in hive that may indicate swarming activity, reduced nectar intake, food consumption, or colony stress. The query also illustrates the interplay between ontology classes, object properties and data properties for analytical reasoning over temporal hive observations as seen in Figure 8.

Listing 3: SPARQL query for Competence Question 3 (CQ3) in Jena Fuseki Server for Ontology Evaluation

```
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
PREFIX xsd: <http://www.w3.org/2001/XMLSchema#>
PREFIX pbo: <http://www.semanticweb.org/paulk/ontologies/2025/10/
    revisedPBO#>
PREFIX sosa: <http://www.w3.org/ns/sosa/>
SELECT ?hive ?day ?startWeight ?endWeight (?endWeight - ?
    startWeight AS ?dailyChange)
WHERE {
    # 1) For each hive + day, get first and last timestamps
    {
        SELECT ?hive ?day (MIN(?t) AS ?startTime) (MAX(?t) AS ?endTime)
        WHERE {
            ?obs rdf:type pbo:WeightObservation ;
                sosa:hasFeatureOfInterest ?hive ;
                pbo:recordedAt ?t .
            # Optional: limit to a specific period
            FILTER(
                ?t >= "2017-01-01T00:00:00"^^xsd:dateTime &&
                ?t < "2017-01-31T00:00:00"^^xsd:dateTime
            )
            # Derive calendar day from timestamp
            BIND( xsd:date(?t) AS ?day )
        }
        GROUP BY ?hive ?day
    }
    # 2) Get weight at startTime
    ?startObs rdf:type pbo:WeightObservation ;
        sosa:hasFeatureOfInterest ?hive ;
        pbo:recordedAt ?startTime ;
```

```

        pbo:hasWeight ?startWeight .
# 3) Get weight at endTime
?endObs    rdf:type pbo:WeightObservation ;
           sosa:hasFeatureOfInterest ?hive ;
           pbo:recordedAt ?endTime ;
           pbo:hasWeight ?endWeight .

# 4) Keep only significant gains (e.g. greater than 0 kg)
FILTER( (?endWeight - ?startWeight) > 0 )
}
ORDER BY ?hive ?day

```

Table 6: SPARQL query results for Competence Question 3 (CQ3): daily hive-weight losses

Hive	Day	Start weight (kg)	End weight (kg)	Daily change (kg)
Schwartau	2017-01-02	50.70	50.61	-0.09
Schwartau	2017-01-04	50.65	50.41	-0.24
Schwartau	2017-01-05	50.17	50.05	-0.12
Würzburg	2017-01-03	52.70	52.63	-0.07
Würzburg	2017-01-05	52.60	52.57	-0.03

6.4 Example Query 4 for CQ4

Example Query 4 evaluates whether the Schwartau hive maintained a stable internal hive temperature over consecutive hourly observations. The query as seen in Listing 4 uses instances of `pbo:InternalTemperatureObservation` class to represent internal temperature of the hive measurements collected from sensors. Each observation is linked to the monitored hive through the object property `sosa:hasFeatureOfInterest` with `pbo:Schwartau` as the feature of interest. The timestamp of each observation is represented by the data property `pbo:recordedAt`, while the measured internal temperature value is represented by the data property `pbo:hasTemperature`.

The query retrieves three consecutive hourly observations represented by `?obs1`, `?obs2` and `?obs3`. Temporal continuity between the observations is established by requiring the second observation to occur one hour after the first, and the third observation to occur one hour after the second. This ensures that each observation is exactly one hour apart from the previous one. The query then evaluates whether at least one of the recorded temperature values is greater than 20°C using the logical OR condition.

As shown in Table 7, the query returned 14 records for sequences of consecutive hourly temperature observations from the Schwartau hive. The results indicate that the hive maintained internal temperatures above the specified threshold during the evaluated observation periods. These findings demonstrate that the developed PBO ontology can support temporal analysis of sensor-based internal temperature observations of the hive through the interaction of ontology classes, object properties and data properties as shown in Figure 9.

Listing 4: SPARQL query for Competence Question 4 (CQ4) in Jena Fuseki Server for Ontology Evaluation

```
PREFIX sosa: <http://www.w3.org/ns/sosa/>
PREFIX rdf:  <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
PREFIX xsd:  <http://www.w3.org/2001/XMLSchema#>
PREFIX pbo:  <http://www.semanticweb.org/paulk/ontologies/2025/10/
    revisedPBO#>
SELECT ?time1 ?temp1 ?time2 ?temp2 ?time3 ?temp3
WHERE {
    # First hour
    ?obs1 a pbo:InternalTemperatureObservation ;
          sosa:hasFeatureOfInterest pbo:Schwartau ;
          pbo:recordedAt ?time1 ;
          pbo:hasTemperature ?temp1 .
    # Second hour = first + 1 hour
    ?obs2 a pbo:InternalTemperatureObservation ;
          sosa:hasFeatureOfInterest pbo:Schwartau ;
          pbo:recordedAt ?time2 ;
          pbo:hasTemperature ?temp2 .
    # Third hour = second + 1 hour
    ?obs3 a pbo:InternalTemperatureObservation ;
          sosa:hasFeatureOfInterest pbo:Schwartau ;
          pbo:recordedAt ?time3 ;
          pbo:hasTemperature ?temp3 .
    FILTER(?time2 = ?time1 + "PT1H"^^xsd:duration)
    FILTER(?time3 = ?time2 + "PT1H"^^xsd:duration)
    # keep cases where at least one of the three hourly readings is
    above 20
    FILTER(?temp1 > 20.0 || ?temp2 > 20.0 || ?temp3 > 20.0)
}
ORDER BY ?time1
```

Table 7: SPARQL query results for Competence Question 4 (CQ4): consecutive internal-temperature observations for the Schwartau hive

Time 1	Temp. 1 (°C)	Time 2	Temp. 2 (°C)	Time 3	Temp. 3 (°C)
2017-01-01 19:15	13.93	2017-01-01 20:15	12.52	2017-01-01 21:15	20.56
2017-01-01 20:15	12.52	2017-01-01 21:15	20.56	2017-01-01 22:15	19.91
2017-01-01 21:15	20.56	2017-01-01 22:15	19.91	2017-01-01 23:15	19.99
2017-01-02 10:15	12.52	2017-01-02 11:15	11.50	2017-01-02 12:15	20.91
2017-01-02 10:15	12.52	2017-01-02 11:15	15.62	2017-01-02 12:15	20.91
2017-01-02 11:15	11.50	2017-01-02 12:15	20.91	2017-01-02 13:15	14.60
2017-01-02 11:15	11.50	2017-01-02 12:15	20.91	2017-01-02 13:15	15.63
2017-01-02 11:15	15.62	2017-01-02 12:15	20.91	2017-01-02 13:15	14.60
2017-01-02 11:15	15.62	2017-01-02 12:15	20.91	2017-01-02 13:15	15.63
2017-01-02 12:15	20.91	2017-01-02 13:15	14.60	2017-01-02 14:15	11.91
2017-01-02 12:15	20.91	2017-01-02 13:15	15.63	2017-01-02 14:15	11.91
2017-01-02 20:15	11.67	2017-01-02 21:15	13.48	2017-01-02 22:15	20.08
2017-01-02 21:15	13.48	2017-01-02 22:15	20.08	2017-01-02 23:15	15.84
2017-01-02 22:15	20.08	2017-01-02 23:15	15.84	2017-01-03 00:15	16.57

6.5 Example Query 5 for CQ5

Example Query 5 evaluates whether the Schwartau hive experienced rapid internal temperature drops greater than 2°C within one hour. The query as seen in Listing 5 uses instances of `pbo:InternalTemperatureObservation` class to represent internal temperature measurements of the hive. Each observation is linked to the Schwartau hive through the object property `sosa:hasFeatureOfInterest`. The timestamp of each observation is represented by the data property `pbo:recordedAt`, while the measured internal temperature value is represented by the data property `pbo:hasTemperature`.

The query compares pairs of consecutive temperature observations that are exactly one hour apart. It computes the temperature difference as the second temperature minus the first temperature. Therefore, a negative value of `deltaTemp` indicates a temperature decrease. The filter condition keeps only cases where this difference is less than -2.0, meaning that the internal temperature decreased by more than 2°C within one hour.

As shown in Table 8, the query returned twelve cases in which the Schwartau hive experienced rapid hourly temperature drops. For example, some returned values show decreases of more than 4°C, 5°C, or even 7°C within one hour. These results demonstrate that the developed PBO ontology can support sensor-based detection of abnormal internal temperature of the hive changes by combining observation classes, object properties and data properties as seen in Figure 9. Such rapid drops may indicate weakening thermoregulation, disturbance, colony stress or other conditions requiring beekeeper attention.

Listing 5: SPARQL query for Competence Question 5 (CQ5) in Jena Fuseki

Server for Ontology Evaluation

```

PREFIX sosa: <http://www.w3.org/ns/sosa/>
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
PREFIX xsd: <http://www.w3.org/2001/XMLSchema#>
PREFIX pbo: <http://www.semanticweb.org/paulk/ontologies/2025/10/
    revisedPBO#>
SELECT ?time1 ?temp1 ?time2 ?temp2 ?deltaTemp
WHERE {
    ?obs1 a pbo:InternalTemperatureObservation ;
        sosa:hasFeatureOfInterest pbo:Schwartau ;
        pbo:recordedAt ?time1 ;
        pbo:hasTemperature ?temp1 .
    ?obs2 a pbo:InternalTemperatureObservation ;
        sosa:hasFeatureOfInterest pbo:Schwartau ;
        pbo:recordedAt ?time2 ;
        pbo:hasTemperature ?temp2 .
    # ensure obs2 is exactly 1 hour after obs1
    FILTER(?time2 = ?time1 + "PT1H"^^xsd:duration)
    # compute temperature change
    BIND((?temp2 - ?temp1) AS ?deltaTemp)
    # keep only strong drops
    FILTER(?deltaTemp < -2.0)
}
ORDER BY ?time1

```

Table 8: SPARQL query results for Competence Question 5 (CQ5): rapid internal-temperature drops greater than 2°C within one hour

Time 1	Temp. 1 (°C)	Time 2	Temp. 2 (°C)	Change (°C)
2017-01-01 17:15	17.63	2017-01-01 18:15	13.35	-4.28
2017-01-02 00:15	18.24	2017-01-02 01:15	13.47	-4.77
2017-01-02 03:15	14.86	2017-01-02 04:15	12.34	-2.52
2017-01-02 07:15	15.16	2017-01-02 08:15	11.72	-3.44
2017-01-02 11:15	15.62	2017-01-02 12:15	12.92	-2.70
2017-01-02 12:15	20.91	2017-01-02 13:15	15.63	-5.28
2017-01-02 12:15	20.91	2017-01-02 13:15	14.60	-6.31
2017-01-02 13:15	14.60	2017-01-02 14:15	11.91	-2.69
2017-01-02 13:15	15.63	2017-01-02 14:15	11.91	-3.72
2017-01-02 22:15	20.08	2017-01-02 23:15	15.84	-4.24
2017-01-03 02:15	19.50	2017-01-03 03:15	12.33	-7.17
2017-01-03 06:15	14.14	2017-01-03 07:15	11.00	-3.14

6.6 Example Query 6 for CQ6

Example Query 6 evaluates whether the internal temperature of the Schwartau hive shows a consistent upward or downward trend across three

consecutive days. The query as seen in Listing 6 uses instances of `pbo:InternalTemperatureObservation` class to represent internal temperature measurements of the hive. Each observation is linked to the Schwartau hive through the object property `sosa:hasFeatureOfInterest`. The timestamp of each observation is represented by the data property `pbo:recordedAt`, while the measured internal temperature value is represented by the data property `pbo:hasTemperature`.

The query calculates the average internal temperature for three selected days using `AVG(?temp)`. It then computes the difference between the average temperature of Day 2 and Day 1, and between Day 3 and Day 2. These two differences represent the temperature trend between consecutive days. A positive value indicates an upward trend, while a negative value indicates a downward trend.

As shown in Table 9, the query returned one result row containing the average temperatures for the three selected days and their corresponding trend values. The trend values are negative, indicating a downward internal temperature trend across the evaluated days. This result demonstrates that the developed PBO ontology can support temporal analysis of sensor-based hive observations by combining classes, object properties and data properties as seen in Figure 9. Such a trend may indicate changes in colony behaviour, possible cluster movement inside the hive, environmental influence, or another condition requiring further beekeeper interpretation.

Listing 6: SPARQL query for Competence Question 6 (CQ6) in Jena Fuseki Server for Ontology Evaluation

```
PREFIX sosa: <http://www.w3.org/ns/sosa/>
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
PREFIX xsd: <http://www.w3.org/2001/XMLSchema#>
PREFIX pbo: <http://www.semanticweb.org/paulk/ontologies/2025/10/
    revisedPBO#>
SELECT ?avgDay1 ?avgDay2 ?avgDay3
(?avgDay2 - ?avgDay1 AS ?trend1_Day2_Day1)
    (?avgDay3 - ?avgDay2 AS ?trend2_Day3_Day2)
WHERE {
    # Day 1: e.g. 2017-01-01
    {
        SELECT (AVG(?temp) AS ?avgDay1)
        WHERE {
            ?obs rdf:type pbo:InternalTemperatureObservation ;
                sosa:hasFeatureOfInterest pbo:Schwartau ;
                pbo:recordedAt ?time ;
                pbo:hasTemperature ?temp .
```

```

        FILTER( xsd:date(?time) = "2017-01-01"^^xsd:date )
    }
}

# Day 2: e.g. 2017-01-02
{
    SELECT (AVG(?temp) AS ?avgDay2)
    WHERE {
        ?obs rdf:type pbo:InternalTemperatureObservation ;
            sosa:hasFeatureOfInterest pbo:Schwartau ;
            pbo:recordedAt ?time ;
            pbo:hasTemperature ?temp .
        FILTER( xsd:date(?time) = "2017-01-02"^^xsd:date )
    }
}

# Day 3: e.g. 2017-01-03
{
    SELECT (AVG(?temp) AS ?avgDay3)
    WHERE {
        ?obs rdf:type pbo:InternalTemperatureObservation ;
            sosa:hasFeatureOfInterest pbo:Schwartau ;
            pbo:recordedAt ?time ;
            pbo:hasTemperature ?temp .
        FILTER( xsd:date(?time) = "2017-01-03"^^xsd:date )
    }
}
}

```

Table 9: SPARQL query results for Competence Question 6 (CQ6): internal-temperature trend across three selected days

Result item	Value
Average temperature on Day 1	15.34°C
Average temperature on Day 2	13.78°C
Average temperature on Day 3	14.49°C
Trend from Day 1 to Day 2	-2.06°C
Trend from Day 2 to Day 3	0.72°C

7 Conclusion

This documentation gives the technical details of the PBO ontology about metadata, reused external links, the structure of the ontology, constraints

specified in PBO ontology and SPARQL queries and results during evaluation. The PBO ontology can be extended to improve interoperability by including external links to the concepts and object properties from other ontologies. Also, PBO ontology can be extended by considering more parameters such as acoustic sound, humidity, CO2 concentration and geo position as the real-world data from hives connected with sensors in the apiary.